Keeping the Spikes at the Edge: Wavelet Approximation of Machine Waveforms at a Tenth of Lossless Size

Version 0.2 — hypotheses, method and development findings

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ABSTRACT

A sensor at a machine produces far more waveform than an edge device can send or keep. Lossless storage needs about 1.5 bytes per sample; a cellular or low-power uplink affords a tenth of that or less. Downsampling and statistics fit the budget but remove the short transients diagnosis looks for, and general-purpose error-bounded compressors are tuned for smooth scientific fields, not for impulsive vibration. We ask which approximation keeps the most of what diagnosis needs at 0.05–0.3 bytes per sample, and whether a small device can produce it in real time. Our candidate is a plain transform coder: an orthogonal wavelet transform, one absolute threshold for all levels, uniform quantisation, varint and deflate coding, with rate control per window by bisection on the threshold. We state four falsifiable hypotheses (error per byte against SZ3, ZFP and decimation; whether diagnosis survives; real time on a Raspberry Pi; whether Bayesian coefficient selection is needed) and report development rounds on bearing, seismic, screwdriving and CNC milling data and on a Raspberry Pi. On development data the coder has the lowest error at nearly every budget, a run-to-failure bearing fault is detected on the same file as on the original down to 0.05 bytes per sample, while SZ3 and decimation detect it 17 and 7 files later at 0.1, a Raspberry Pi 3B+ encodes at twice real time for 64 kHz on one core, and a heavy-tailed Bayesian prior adds nothing. It also raises false seismic triggers 2.4-fold at 0.1 bytes per sample, which the confirmation will test. The study (F2) will be frozen before its confirmation data are used.

1 INTRODUCTION

Condition monitoring at the edge faces a budget at every step. A 64 kHz vibration channel produces 5.5 billion samples a day; lossless storage in MorletDB needs 1.5–1.7 bytes per sample [34], about 9 GB a day, and a cellular or LoRa uplink carries a small fraction of that. What diagnosis needs from the signal is often short: an impact train in the envelope spectrum [4, 36], the onset of a seismic arrival [3, 44], the shape of a tightening curve. Downsampling removes the high-frequency content that carries impacts; statistics remove the shape. Error-bounded lossy compressors [16, 26, 27] bound the error per sample, which for broadband vibration spends the budget on noise.

Wavelets concentrate transients in few large coefficients and spread stationary background over many small ones [15, 28]; keeping the largest coefficients is the best approximation for their number in an orthogonal basis, and wavelet compression of vibration data has been used offline [40] and on sensor nodes [45]. What is missing is a measurement of what such an approximation preserves for diagnosis, against today's error-bounded compressors, at the budgets an edge uplink affords, and on a device of the size that would run it.

Contributions. (1) A small transform coder with per-window rate control that an edge device can run. (2) Four falsifiable hypotheses with refutation criteria, fixed before confirmation data are used. (3) Development findings on bearing, seismic and screwdriving data, including the mistakes that changed the method. (4) A place in a waveform database: the approximation as a level between exact samples and summaries in a graded-forgetting ladder [33].

2 METHOD

A window x (one second of vibration, one screwdriving cycle, 2^{15} seismic samples) is decomposed with an orthogonal discrete wavelet transform (sym8; 6, 4 and 8 levels). Every detail coefficient with $|d| > T$, $T = c \cdot \mathrm{std}(x)$, is kept; kept coefficients and the coarsest band are quantised with step qT. The positions of kept coefficients (as gaps) and their values (zigzag) are written as LEB128 varints, preceded by the length, the number of levels and the step, and the stream is compressed with deflate. Decoding inverts each step. *Rate control*: for a budget B in bytes per sample, the device bisects on c (eight tries, $q = 0.5$, the transform computed once) and sends the smallest threshold whose code fits B. The development harness used a fixed grid of (c, q) instead, choosing the smallest error within B; on the device, bisection was both faster and slightly better (Section 4).

Comparisons, each with the same rate control: SZ3 [26] and ZFP [27] at absolute error bounds; decimation (FIR, zero phase) with quantised, delta-coded values and the same coder; and the same coder with coefficients selected by the posterior of an empirical-Bayes spike-and-slab model with a heavy-tailed slab [1, 22, 37].

3 HYPOTHESES

H1, error per byte. At 0.05, 0.1, 0.2 and 0.3 bytes per sample the wavelet coder has lower median relative error than SZ3, ZFP and decimation on every dataset. *Refuted* at a budget if SZ3 or ZFP is as good or better on two of three datasets.

H2, diagnosis survives. At 0.1 bytes per sample, analysis on the approximation finds what it finds on the original: a run-to-failure bearing fault is detected in the envelope spectrum no more than 1% of the run's files later; STA/LTA triggers on at least 95% of the catalogued arrivals the original triggers on, with at most 10% more triggers elsewhere; kurtosis separates failed screwdriving cycles, and a label-free score of RMS and kurtosis separates bad CNC milling runs, with an AUROC within 0.02 of the original. *Refuted* if any fails.

H3, real time on a small device. On a Raspberry Pi 3B+ the encoder keeps up with a 64 kHz channel on one core with less than 64 MB of memory. *Refuted* if not.

Table 1: Development, round 3: median relative error at 0.1 bytes per sample, and diagnosis on the approximation at 0.1 (original in brackets).

	Wavelet	SZ3	Decimation
Error, CWRU	**0.56**	0.93	0.86
Error, IMS	**0.70**	0.95	0.74
Error, seismic	**0.038**	0.063	0.079
Error, PyScrew	0.090	–	**0.086**
Error, CNC milling	**0.42**	0.79	0.47
IMS fault detected at file (532)	**532**	549	539
Seismic arrivals triggered (28 of 30)	29	26	26
Seismic triggers elsewhere (400)	955	**393**	556
PyScrew kurtosis AUROC (0.961)	0.957	–	0.954
CNC bad-run AUROC (0.765)	0.745	0.717	0.737
CNC RMS deviation	**4.2%**	5.4%	9.3%

H4, no Bayesian machinery needed. Posterior selection does not lower the error by more than 5% against the plain threshold at any budget. *Refuted*, and the Bayesian arm recommended, if it does on two of three datasets.

4 DEVELOPMENT FINDINGS

Development used CWRU bearings, IMS set 2 (a bearing run to an outer-race failure), PyScrew scenario s01, two weeks of four seismic stations with a public catalogue and one of three machines of the Bosch CNC milling data [41], all separate from the confirmation data [31]. The first version of the study also proposed a label-free relevance score from the same posterior; it was not better than kurtosis on development data and was dropped before any freeze.

What changed the method. A threshold set from a noise estimate per level, the usual choice for denoising, treated the seismic microseism as noise and removed most of the signal; a coder needs one absolute threshold for all levels. STA/LTA on the raw trace triggered thousands of times on the original itself and needs a band-pass (2–20 Hz). Broadband vibration needs thresholds well above the signal's standard deviation to reach 0.05 bytes per sample. Fidelity of a diagnostic measure is two-sided: reconstructions can inflate a fault peak. And the posterior mean of a Bayesian model, a good denoiser, is a source of error in an archive.

Table 1 summarises the third round. The wavelet coder has the lowest error on bearing and seismic data at every budget from 0.1 up (and at 0.05 except on IMS, where decimation is marginally better), and ties decimation on short screwdriving cycles; SZ3 is never better below 0.3 bytes per sample, and ZFP's one-dimensional error-bound mode is not competitive. The bearing fault is detected on the same file as on the original at every budget down to 0.05; SZ3 detects it 172 files later at 0.05 and 17 at 0.1. The weak point is seismic false triggers: thresholding leaves artefacts that STA/LTA reads as onsets, 2.4 times as many triggers away from catalogued arrivals at 0.1 and 12% more at 0.3, so H2 as stated fails on development data at 0.1. Posterior selection was worse than the plain threshold (0.57 against 0.56 at 0.1 and 0.45 against 0.40 at 0.2 on CWRU), so H4 holds on development data. On CNC milling the coder again has the lowest error, bad runs remain about as separable as on the original (AUROC 0.745 against 0.765, at the edge of H2's 0.02), and per-window RMS, the usual wear proxy, deviates by 4% at 0.1 and 1.7% at 0.2; on a machine tool the approximation should travel with exact per-window statistics, which MorletDB already keeps at about 1% of the lossless size. On the Raspberry Pi 3B+ of our earlier edge study, one core of the Python prototype encodes at 13 times real time for 64 kHz with one fixed setting and at 2.1 times with bisection rate control, within 56 MB; the development harness's grid search (56 tries) would not keep up (0.12 times), and bisection, with a continuous threshold, reached lower error (0.67 against 0.71) at the same budget. We report these before freezing so that the confirmation cannot be tuned to them.

5 RELATED WORK

Wavelet thresholding and its Bayesian forms. Thresholding rules [17, 18, 21] and their Bayesian counterparts with spike-and-slab priors [1, 13, 14, 42, 43], including empirical Bayes with level-wise weights [22–24], heavy-tailed and Pareto-based priors [2, 8, 19, 37, 38] and shrinkage priors outside wavelets [5, 9, 10, 39], target denoising; BayesShrink links shrinkage to compression [12]. Our development data show that for an archive the plain threshold suffices.

Compression. Error-bounded lossy compressors [16, 26, 27] and compressors for time series [11, 32] bound the error per value; wavelet compression of vibration data was studied offline [40] and on wireless nodes [45], and compressive sensing [7, 20] moves the work to the decoder.

Edge data reduction. Send-on-delta and event-triggered sampling [30], edge reduction for condition monitoring [29, 35] and tiny models on microcontrollers [6, 25] decide what to send; we measure what an approximation keeps when everything is sent at a fixed budget.

6 LIMITATIONS

This version states hypotheses and development findings, not confirmed results. Development used one bearing test rig, one run-to-failure bearing, one screwdriving scenario and two weeks of four seismic stations. Relative error and three diagnostic proxies stand in for an engineer's judgement. The encoder is a Python prototype; real-time operation on the device is not yet measured.

USE OF AI ASSISTANCE AND RESPONSIBILITY

Code, experiment harnesses, analyses and text were developed with Claude (Anthropic), an AI assistant, working under the author's direction. The author reviewed the content and is responsible for it.

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